PHASE 4.2.2 TECHNICAL & GOVERNANCE SANITY AUDIT

SimuPharm Educational Pilot Evaluation

Final Pre-Recruitment Statistical Verification, Schema Validation & Governance Lock

Date: October 4, 2026
Status: NO-GO — ETHICS PENDING
Engine State: PASS — ENGINE FROZEN

1. EXECUTIVE SUMMARY & AUDIT SCOPE

This technical audit constitutes the definitive sanity check of the pre-recruitment protocol package for the **SimuPharm Educational Evaluation Study**. In strict compliance with the Phase 4 charter, **no study redesign has been performed, no simulation source code has been altered, no participant-facing features or arms have been added, and no student data has been simulated**. The audit examined all statistical code implementations, non-parametric effect sizes, confidence interval mathematics, data dictionary schemas, multiple-comparison hierarchies, 50-minute session logistics, and publication claims across the repository.

> **Key Audit Outcome:** All computational implementations (paired Pratt Wilcoxon, signed rank-biserial correlation, Walsh-average order-statistic confidence intervals, and complete-case filtering) have been verified across 7 stress-test edge cases. All overreaching claim language and unsupported journal compliance declarations have been systematically purged. The protocol package is methodologically and statistically sealed.

2. ISSUES IDENTIFIED & CORRECTIONS IMPLEMENTED

#	Audit Domain	Issue Identified in Frozen Package	Corrective Action Implemented	Status
1	**Analysis Terminology**	Primary outcome labeled as "Primary Confirmatory Analysis", inappropriately implying definitive evidence of educational effectiveness for an exploratory pilot ($N \approx 25$).	Replaced with **"Primary Prespecified Inferential Analysis"** across all research questions, SAP, and analysis plans. Confirmatory framing removed.	RESOLVED
2	**Wilcoxon SciPy Edge Case**	SciPy's `stats.wilcoxon(..., zero_method='pratt')` raises an unhandled `ValueError` when all paired differences are zero ($\Delta\text{Score} = 0 \ \forall i$).	Implemented an explicit all-zero difference safeguard: catches zero-difference arrays and returns $W = 0$, $p = 1.0$, $r_{rb} = 0.0$, and Hodges-Lehmann difference $= 0.0$.	RESOLVED
3	**Effect-Size Direction Flaw**	Formula $r = 1 - \frac{2W}{S}$ using SciPy's test statistic $W = \min(W^+, W^-)$ flips signs and yields $+1.0$ for both all-positive and all-negative differences.	Replaced with the exact signed rank-biserial formula: $r_{rb} = \frac{W^+ - W^-}{W^+ + W^-}$. Positive indicates improvement, negative indicates deterioration, zero indicates null effect.	RESOLVED
4	**Hodges–Lehmann CI Audit**	The protocol stated a "95% BCa bootstrap confidence interval" for Hodges–Lehmann, but it was unbuilt in code and mathematically prone to jackknife instability for discrete scores.	Implemented the exact **Walsh-average order-statistic confidence interval (Bauer 1972; Hollander & Wolfe 1999)**, which directly inverts the Wilcoxon signed-rank test.	RESOLVED
5	**Confidence-Calibration Scope**	Certain sections previously referenced confidence analysis as demonstrating "calibrated metacognition" or "simulator-induced calibration".	Re-anchored strictly as an **exploratory confidence–performance association indicator** (Spearman ρ, Pearson r). All causal claims excised.	RESOLVED
6	**Expert Review Thresholds**	Lynn (1986) cutoffs ($\text{I-CVI} \ge 0.80$, $\text{S-CVI/Ave} \ge 0.90$) were presented as universal mandatory hurdles rather than descriptive panel-size benchmarks.	Specified exact panel size (3–5 experts), formulas, and noted that small N requires unanimous consensus ($3/3$) or $4/5$; characterized as descriptive benchmarks.	RESOLVED
7	**Open-Science Compliance Claim**	Claimed all assets conform to "JCE Transparency and Openness Promotion (TOP) Guidelines Level 2" without external journal verification.	Excised TOP Level 2 claim; replaced with: "*Open-science practices are structured to support transparency, reproducibility, and responsible data/code sharing.*"	RESOLVED
8	**Multiple-Testing Family**	All 5 secondary metrics (Rubric, Latency, Calibration, SUS, Hake) were lumped into a single Holm family, combining hypothesis tests with descriptive indices.	Restructured into a 3-tier hierarchy: (1) Primary Prespecified Inferential; (2) Secondary Inferential (SUS, Rubric; Holm $k=2$); (3) Exploratory & Descriptive.	RESOLVED
9	**50-Minute Lab Feasibility**	Answering 3 extensive qualitative reflection prompts alongside 10 post-test items, 10 SUS items, and 8 UX items in 10 min risked severe cognitive strain.	Shortened qualitative instructions: explicitly designated as optional bullet-point/short-phrase feedback (1–2 sentences, ≤ 30 words), preserving the 50-min ceiling.	RESOLVED
10	**Answer Key Claim Wording**	Answer key rationale for Post-Test Q4 asserted that peak area scaling "proves contamination".	Calibrated to "indicates contaminants accumulate from weak mobile phase component during equilibration."	RESOLVED

3. STATISTICAL IMPLEMENTATION VERIFICATION (DEMO_TEST_ONLY SUITE)

A dedicated technical verification script (`verify_wilcoxon_pipeline.py`) was executed across seven synthetic stress-test datasets representing mathematical boundary conditions. No participant data was generated or analyzed:

Dataset Case	Synthetic Paired Scores	Wilcoxon Pratt Output	Signed r_{rb}	Hodges–Lehmann & 95% CI	Technical Verification
Case A: All Positive	Pre: [2, 3, 4, 3, 2, 4, 3, 2] Post: [5, 6, 7, 6, 5, 8, 7, 6] Δ: [+3, +3, +3, +3, +3, +4, +4, +4]	$W^+ = 36.0, \ W^- = 0.0$ $W = 0.0, \ p = 0.00781$	**+1.000**	HL = +3.50 95% Walsh: [+3.00, +4.00]	PASS: STRICT POSITIVE
Case B: All Negative	Pre: [6, 7, 8, 7, 6, 8, 7, 6] Post: [3, 4, 5, 4, 3, 5, 4, 3] Δ: [-3, -3, -3, -3, -3, -3, -3, -3]	$W^+ = 0.0, \ W^- = 36.0$ $W = 0.0, \ p = 0.00781$	**-1.000**	HL = -3.00 95% Walsh: [-3.00, -3.00]	PASS: STRICT NEGATIVE
Case C: Mixed Differences	Pre: [3, 4, 5, 6, 4, 5, 3, 6] Post: [6, 5, 4, 7, 6, 4, 5, 8] Δ: [+3, +1, -1, +1, +2, -1, +2, +2]	$W^+ = 31.0, \ W^- = 5.0$ $W = 5.0, \ p = 0.07813$	**+0.722**	HL = +1.25 95% Walsh: [0.00, +2.50]	PASS: PROPORTIONAL RANK
Case D: Zeros Present	Pre: [4, 5, 5, 6, 3, 4, 5, 5] Post: [6, 5, 7, 6, 5, 4, 8, 5] Δ: [+2, 0, +2, 0, +2, 0, +3, 0]	$W^+ = 26.0, \ W^- = 0.0$ $W = 0.0, \ p = 0.04743$	**+1.000**	HL = +1.00 95% Walsh: [0.00, +2.50]	PASS: PRATT ZEROS PRESERVED
Case E: All Zeros	Pre: [5, 5, 5, 5, 5] Post: [5, 5, 5, 5, 5] Δ: [0, 0, 0, 0, 0]	$W^+ = 0.0, \ W^- = 0.0$ $W = 0.0, \ p = 1.00000$	**0.000**	HL = 0.00 95% Walsh: [0.00, 0.00]	PASS: CRASH SAFEGUARD

Dataset Case	Synthetic Paired Scores	Wilcoxon Pratt Output	Signed r_{rb}	Hodges–Lehmann & 95% CI	Technical Verification
Case F: Balanced Ties	Pre: [3, 3, 3, 3, 3, 3] Post: [5, 5, 5, 1, 1, 1] Δ: [+2, +2, +2, -2, -2, -2]	$W^+ = 10.5, \ W^- = 10.5$ $W = 10.5, \ p = 1.00000$	**0.000**	HL = 0.00 95% Walsh: [-2.00, +2.00]	**PASS: AVERAGE TIES NULL**
Case G: Missing Values	Pre: [3, 4, NaN, 5, 6, NaN, 4] Post: [5, NaN, 6, 7, 6, NaN, 7] Pairs: 4 valid, 3 excluded	$W^+ = 9.0, \ W^- = 0.0$ $W = 0.0, \ p = 0.09182$	**+1.000**	HL = +2.00 95% Walsh: [0.00, +3.00]	**PASS: COMPLETE-CASE N=4**

4. HODGES–LEHMANN & EFFECT-SIZE METHODOLOGICAL VERIFICATION

Hodges–Lehmann Walsh CI Specification

For N paired observations, $M = \frac{N(N+1)}{2}$ pairwise Walsh averages $A_{(k)} = \frac{d_i + d_j}{2}$ ($1 \le i \le j \le N$) are calculated and sorted in ascending order.

- **Point Estimate:** $\text{HL} = \text{Median}(\{A_{(k)}\})$.
- **Exact Inversion CI (Bauer 1972):** The $100(1-\alpha)\%$ confidence bounds are defined by order statistics $A_{(k_{\text{crit}})}$ and $A_{(M - k_{\text{crit}} + 1)}$, where $k_{\text{crit}} = \left\lfloor \frac{N(N+1)}{4} - z_{1-\alpha/2}\sqrt{\frac{N(N+1)(2N+1)}{24}} \right\rfloor$.
- **Defensibility:** For $N=25$, $M = 325$, $k_{\text{crit}} = 89$. Exact CI is $[A_{(89)}, A_{(237)}]$. Eliminates bootstrap instability on discrete integers.

Signed Rank-Biserial Effect Size (r_{rb})

The Kerby (2014) rank-biserial correlation is computed directly from positive and negative rank sums under Pratt's ranking:

- **Formula:** $r_{rb} = \frac{W^+ - W^-}{W^+ + W^-} \quad (\text{if } W^+ + W^- = 0, \ r_{rb} = 0.0)$.
- **Directional Integrity:**
  - $r_{rb} > 0$: cohort improvement (post-test > pre-test).
  - $r_{rb} < 0$: cohort deterioration (post-test < pre-test).
  - $r_{rb} = 0$: no directional shift (balanced ranks).
- **Manuscript Guidance:** Prevents sign errors that arise when using $\min(W^+, W^-)$ in unadjusted formulas.

5. 73-COLUMN DATA-SCHEMA EXHAUSTIVE AUDIT

Every column of the 73-variable schema (`participant_data_template.csv`) was audited against the SAP, instruments, and analysis scripts:

Column Variable(s)	Measurement Purpose	Instrument Source	Analysis Pipeline Utilization	Required?	Audit Decision
`participant_id`	Unique anonymous tracking code	Envelope Code	Record integrity, deduplication	YES	**RETAINED**
`degree_program`	Student major (B.Pharm, Chem)	Demographic Form	Exploratory subgroup descriptive summary	YES	**RETAINED**
`prior_hplc_lab`	Prior wet-lab experience (0/1)	Demographic Form	Exploratory subgroup descriptive summary	YES	**RETAINED**
`pre_q1` to `pre_q10` (10 cols)	Item-level pre-test accuracy (0/1)	Pre-Test Form	Exploratory item analysis (P, D), Cronbach's α	YES	**RETAINED**
`pre_test_total`	Pre-test total score (0–10)	Calculated	Wilcoxon signed-rank, paired t-test, Hake gain	YES	**RETAINED**
`post_q1` to `post_q10` (10 cols)	Item-level post-test accuracy (0/1)	Post-Test Form	Exploratory item analysis (P, D), Cronbach's α	YES	**RETAINED**
`post_test_total`	Post-test total score (0–10)	Calculated	Wilcoxon signed-rank, paired t-test, Hake gain	YES	**RETAINED**
`score_gain`	Paired gain ($\text{Post} - \text{Pre}$)	Derived	Primary outcome distribution & Shapiro-Wilk	Derived	**RETAINED (CONV)**
`hake_gain`	Normalized gain $\langle g \rangle$	Derived	Descriptive gain categorization (low/med/high)	Derived	**RETAINED (CONV)**
`task_alpha_id_score`, `task_alpha_reason_score`, `task_alpha_action_score`	Scenario Alpha component rubric scores (0, 1, 2 each)	Rubric Sheet	Component-level descriptive reporting (Fitting Leak)	YES	**RETAINED**
`task_alpha_rubric_total`	Alpha scenario total (0–6)	Derived	Task difficulty comparison	Derived	**RETAINED (CONV)**
`task_alpha_time_s`	Alpha diagnostic latency (s)	Worksheet Stamp	Exploratory diagnostic efficiency (median/IQR)	YES	**RETAINED**
`task_alpha_conf`	Alpha subjective confidence (1–5)	Worksheet Scale	Exploratory confidence–accuracy correlation	YES	**RETAINED**
Tasks Beta, Gamma, Delta (18 columns total)	Corresponding 6 metrics for Voids, Air Bubbles, and Contamination	Rubric Sheet & Worksheet	Multi-scenario rubric aggregation, latency distribution, confidence association	YES	**RETAINED**
`rater1_grand_total`, `rater2_grand_total`	Independent grand totals (0–24)	Faculty Raters	Inter-rater reliability: Weighted Kappa (κ_w), $\text{ICC}(2,1)$	YES	**RETAINED**
`task_rubric_grand_total`	Adjudicated grand score (0–24)	Rater Consensus	Secondary inferential test against guessing (6.0)	YES	**RETAINED**
`sus_q1` to `sus_q10` (10 cols)	Canonical Brooke SUS items (1–5)	SUS Instrument	Item profiling, Sauro missing imputation	YES	**RETAINED**
`sus_total_score`	Composite SUS score (0–100)	Derived	Secondary inferential test against 68.0 benchmark	Derived	**RETAINED (CONV)**
`ux_u1` to `ux_u8` (8 cols)	Pedagogical UX items (1–5)	UX Instrument	Formative UI feature diagnostics (medians/IQRs)	YES	**RETAINED**

Schema Audit Summary: Exactly 73 columns. Zero unneeded or phantom variables. 64 primary collected metrics + 9 standard convenience/derived columns (`_total` , `_gain`). Template purity verified: exactly 1 header row, **zero data rows**.

6. MULTIPLE-COMPARISON INFERENTIAL HIERARCHY

To prevent post-hoc p-value fishing and preserve family-wise error rate ($\alpha = 0.05$), the study outcomes are partitioned into three immutable, pre-specified tiers:

TIER 1: PRIMARY PRESPECIFIED INFERENTIAL ANALYSIS (STANDALONE $\alpha = 0.05$)

• **Paired Troubleshooting Score Gain ($\Delta\text{Score} = \text{PostTest} - \text{PreTest}$):** Two-tailed Wilcoxon Signed-Rank Test using Pratt's method (`zero_method='pratt'`). Reported with Hodges–Lehmann median difference (95% Walsh order-statistic CI) and signed rank-biserial correlation (r_{rb}). Paired Student's t-test reported as pre-specified sensitivity analysis.

TIER 2: SECONDARY INFERENTIAL HYPOTHESIS FAMILY (HOLM-BONFERRONI CONTROLLED, $k = 2$)

• **System Usability Acceptability ($H_{0,1}$):** One-sample Wilcoxon signed-rank test comparing SUS composite scores against the industry average benchmark of 68.0 ($H_0: \text{Median SUS} \le 68.0$).

• **Practical Diagnostic Competence ($H_{0,2}$):** One-sample Wilcoxon signed-rank test comparing grand diagnostic rubric scores against the random-guessing threshold ($H_0: \text{Median Rubric} \le 6.0/24$).

Multiplicity Adjustment: Holm-Bonferroni step-down correction: sort $p_{(1)} \le p_{(2)}$. Reject $H_{0,(1)}$ if $p_{(1)} \le 0.025$; reject $H_{0,(2)}$ if $p_{(2)} \le 0.050$.

TIER 3: EXPLORATORY & DESCRIPTIVE OUTCOMES (NO MULTIPLICITY PENALTY, HYPOTHESIS-GENERATING)

• **Hake Normalized Gain ($\langle g \rangle$):** Reported descriptively via Hake's brackets (low < 0.3, med 0.3–0.7, high > 0.7). Identical null to Tier 1.
• **Exploratory Confidence–Performance Association:** Spearman ρ and Pearson r between confidence and rubric performance (zero causal claims).
• **Diagnostic Latency:** Median completion time in seconds with IQR across practical scenarios.
• **Exploratory Item Analysis:** Item difficulty (P) and discrimination (D) with small-sample ($N \approx 25$) caveats.
• **Subgroup Summaries:** B.Pharm vs Chemistry; prior HPLC experience vs novices (medians/IQRs only; zero inferential claims).

7. 50-MINUTE SESSION FEASIBILITY & COGNITIVE LOAD AUDIT

Session timing was audited to ensure strict adherence to standard undergraduate laboratory periods:

Block / Time	Activity Description	Participant Task & Cognitive Burden	Duration Audit	Feasibility
Block 1 $t = 00\text{--}05$ min	Briefing & Consent	Institutional consent form review, voluntary signature, anonymous ID code assignment. Neutral script.	5 minutes	VERIFIED
Block 2 $t = 05\text{--}15$ min	Pre-Test Diagnostic Assessment	10 multiple-choice / matching diagnostic items. Focus on baseline chromatographic problem-solving.	10 minutes (60 s / item)	VERIFIED
Block 3 $t = 15\text{--}27$ min	SimuPharm Active Exploration	Unguided simulation familiarization: flow rate, mobile phase, temperature, and fault injection exploration.	12 minutes	VERIFIED
Block 4 $t = 27\text{--}39$ min	Practical Troubleshooting Tasks	4 blinded simulated scenarios (Alpha: Leak, Beta: Void, Gamma: Bubble, Delta: Contamination). Worksheet recording.	12 minutes (3 min / task)	VERIFIED
Block 5 $t = 39\text{--}49$ min	Post-Test, SUS & Feedback	10 isomorphic post-test items (~5 min), canonical Brooke SUS (~2 min), 8 UX items (~1.5 min), brief qualitative prompts (~1.5 min).	10 minutes	VERIFIED
Block 6 $t = 49\text{--}50$ min	Instrument Debrief & Collection	Submission of envelope packets, verification of completion, participant dismissal.	1 minute	VERIFIED

Cognitive Load Mitigation in Block 5: To ensure students comfortably complete all instruments within 10 minutes, qualitative instructions are explicitly calibrated: short phrases or bullet points (1–2 sentences, ≤ 30 words per prompt) are requested. Zero lengthy essay prompts. Total elapsed session time is locked at exactly **50 minutes**.

8. EXPERT CONTENT-VALIDITY METHODOLOGY AUDIT

Reviewer Panel & Formulas

- **Panel Composition:** 3 to 5 independent analytical chromatography faculty and pharmaceutical educators.
- **Item Content Validity Index:** $\text{I-CVI} = \frac{N_{3,4}}{N_{\text{experts}}}$, where $N_{3,4}$ is the number of experts rating relevance/clarity as 3 (Quite relevant) or 4 (Highly relevant).
- **Scale Content Validity Index:** $\text{S-CVI/Ave} = \frac{1}{K} \sum_{k=1}^K \text{I-CVI}_k$.
- **Status:** Form remains strictly unpopulated. Zero fabricated ratings.

Descriptive Framework & Small-N Caveat

- **Descriptive Benchmarks:** Lynn (1986) cutoffs ($\text{I-CVI} \ge 0.80$, $\text{S-CVI/Ave} \ge 0.90$) are descriptive guidelines, not universal mandatory hurdles.
- **Small-Panel Sensitivity:** When $N_{\text{experts}} = 3$, 100% agreement ($3/3 = 1.00$) is required; a single dissenter drops I-CVI to 0.67. For $N_{\text{experts}} = 5$, $4/5 = 0.80$.
- **Scope Boundary:** Evaluates curricular appropriateness and symptom plausibility; does **NOT** substitute for physical instrument validation.

9. CLAIM-LANGUAGE AUDIT REGISTER (12 PRESCRIBED PHRASES)

All Phase 4.2.1 text files and codebases were scanned for 12 sensitive claim terms. Each occurrence was audited against empirical evidence:

Audited Term	Occurrences	Audit Classification	Context in Phase 4 Artifacts	Action Taken
"proves"	5 matches	NEEDS QUALIFICATION	1 in Answer Key Q4 ("proves contamination"); others in Prohibited Claims ledger.	Replaced with "indicates contamination" in answer key. Preserved in Prohibited register.
"validated"	3 matches	SAFE	Appears in Prohibited Claims register ("Validated across 300+ APIs") and as a limitation warning.	Preserved inside Prohibited Claims register. No unsupported positive claims exist.
"confirmatory"	11 matches	NEEDS QUALIFICATION	Used in RQ1, SAP, and outline ("Primary Confirmatory Analysis").	Replaced with "Primary Prespecified Inferential Analysis" across all files.
"psychometrically sound"	1 match	SAFE	Recorded in Readiness Checklist as an overstatement that was excised.	Preserved as audit record of removal.
"experimentally validated"	1 match	SAFE	Appears in Prohibited Claims ledger ("Experimentally Validated Physical Pressure Dynamics").	Preserved as a prohibited claim.

Audited Term	Occurrences	Audit Classification	Context in Phase 4 Artifacts	Action Taken
"physical validation"	1 match	SAFE	Recorded in Checklist confirming that expert review is NOT claimed as physical validation.	Preserved as governance qualification.
"causal"	8 matches	SAFE	Used exclusively to disclaim causality ("Zero causal claims are made") or in Prohibited Claims.	Preserved as essential methodological disclaimers.
"effective"	0 matches	SAFE	Zero unverified claims of "effectiveness" found in active text.	None required.
"superior"	0 matches	SAFE	Zero claims of superiority over traditional instruction.	None required.
"TOP Level 2"	4 matches	REMOVE	Asserted conformance to "JCE TOP Guidelines Level 2" without external journal verification.	Replaced with generic responsible open-science and reproducibility language.
"generalizable"	0 matches	SAFE	Single-institution $N \approx 25$ pilot explicitly acknowledges non-generalizability.	None required.
"metacognitive improvement"	0 matches	SAFE	Causal metacognitive learning claims previously excised; only association indicator used.	None required.

10. FROZEN-ENGINE INTEGRITY CHECK & RECRUITMENT GATE

Cryptographic verification confirms that the SimuPharm scientific simulation engine remains 100% frozen:

Engine File	Calculated SHA-256 Checksum	Phase 4 Baseline	Audit Verdict
`hplc_simulator_script.js`	`2fad9547f9fc7c671176346e78732be30246a515853631dede21ae8f7d3b7167`	Identical	MATCH (FROZEN)
`script.js`	`9812fea221fee06391a35d2680c28ce375ca62bc26bc265ed07f36d4d96f0600`	Identical	MATCH (FROZEN)
`index.html`	`dfb896b41790efcc365a3721b8ccc7aa8f49861485a74b02538eb38cadbdee36`	Identical	MATCH (FROZEN)
`hplc_simulator.html`	`02b2b226bf291c2466c23832382e296523cecc68811afe26ba65822197afd6bb`	Identical	MATCH (FROZEN)

FINAL RECRUITMENT READINESS VERDICT: NO-GO — INSTITUTIONAL ETHICS AUTHORIZATION PENDING

Gate 1 (Scientific & Statistical Readiness): CLEARED (100%) — All code, assessments, rubrics, analysis pipelines, and templates are technically verified.

Gate 2 (Institutional Ethics Authorization): PENDING — Written determination from the host institution's Institutional Ethics Committee (IEC / IRB) has not yet been issued.

Rule of Law: In accordance with international research ethics standards (Declaration of Helsinki) and institutional governance, no student participant may be recruited, consented, or assessed until formal written authorization is secured. Zero self-exemption is permitted.

PROTOCOL STATUS:

FROZEN PENDING INSTITUTIONAL ETHICS AUTHORIZATION

NEXT SCIENTIFIC ACTION:

REAL PARTICIPANT DATA COLLECTION AFTER AUTHORIZATION